

Web and Multimedia Technologies

Lab class 5 – HTML5 + CSS3

Instructions to use the laboratory computers (room B2/B4):

1. If the computer is off, start it with Windows (all computers have a Linux-Windows dual boot)
2. Your username is *cl* followed by the six digits of your student ID; your password is *inf* followed by the last two digits of your student ID (e.g., if the student ID is 123456, the username is *cl123456* and the password is *inf56*)
3. At the end of the class, please logout

Steps:

1. In a folder of your choice, create the following HTML5 file (name it `index.html`):

```
<!DOCTYPE html>
<html>
  <head>
    <meta charset="utf-8">
    <title>Pavia online</title>
    <link rel="stylesheet" href="style.css">
  </head>
  <body>
    <section id="container">
      <header id="header">
        <h1>City of Pavia</h1>
      </header>
      <nav id="menu">
        <a href="#">Home</a>
        <a href="#">Who we are</a>
        <a href="#">Contact us</a>
      </nav>
      <article id="main">
        <h2>Introduction</h2>
        <br>
        
        <p>
          Pavia is the capital of a fertile province known for
          its university and agricultural products including wine,
          rice, cereals, and dairy products.
        </p>
        <video id="myv" src="unipv.mp4" width="720" controls>
          Video tag not supported
        </video><br><br>
        <audio src="bach.mp3" controls>
          Audio tag not supported
        </audio>
      </article>
      <aside id="sidebar">
        <h3>Tourist attractions</h3>
        <ul>
          <li><a href="#">Duomo</a></li>
          <li><a href="#">Castello Visconteo</a></li>
          <li><a href="#">San Pietro in Ciel d'Oro</a></li>
          <li><a href="#">Certosa</a></li>
        </ul>
      </aside>
```

```

        <footer id="footer">
            <p style="margin-left: 1em">
                &copy; All rights reserved - 2025
            </p>
        </footer>
    </section>
</body>
</html>

```

2. Download the three files <https://vision.unipv.it/wmt/lab/html5/bach.mp3>, <https://vision.unipv.it/wmt/lab/html5/pavia.jpg>, and <https://vision.unipv.it/wmt/lab/html5/unipv.mp4> to the same folder where there is `index.html`. Open `index.html` in a web browser and check the result
3. Download the file <https://vision.unipv.it/wmt/lab/html5/style.css> to the same folder where there is `index.html`. After analyzing its content, open `index.html` in a web browser and check the result
4. At the end of the file `style.css`, add the following content and check the result (the page will be *responsive*):

```

/** Media Queries */

@media (min-width: 768px) and (max-width: 980px) {
    #container { margin: 0px auto; }
    #container, #header, #menu, #footer { width: 748px; }
    #main { width: 510px; }
    #sidebar { width: 210px; }
    #myv { width: 500px; }
}

@media (max-width: 767px) {
    #container { margin: 0px auto; }
    #container, #header, #menu, #main, #sidebar, #footer, #myv {
        width: 100%;
    }
}

@media (min-width: 1200px) {
    #container, #header, #menu, #footer { width: 1170px; }
    #main { width: 800px; }
    #sidebar { width: 350px; }
}

```

Add `<meta name="viewport" content="width=device-width, initial-scale=1">` to the `<head>` section of `index.html` (so that it can be displayed properly by mobile devices)

5. In the HTML file, substitute the content of the `<nav>` tag with the following list:

```

<ul>
    <li>
        <a href="">Menu 1</a>
        <ul>
            <li><a href="">Menu 11</a></li>
            <li><a href="">Menu 12</a></li>
            <li><a href="">Menu 13</a></li>
        </ul>
    </li>

```

```

<li>
  <a href="">Menu 2</a>
  <ul>
    <li><a href="">Menu 21</a></li>
    <li>
      <a href="">Menu 22</a>
      <ul>
        <li><a href="">Menu 221</a></li>
        <li><a href="">Menu 222</a></li>
        <li><a href="">Menu 223</a></li>
      </ul>
    </li>
  </ul>
</li>
</ul>

```

6. Download the file <https://vision.unipv.it/wmt/lab/html5/menu.css> to the same folder where there is `index.html` and analyze its content. Load `menu.css` in `index.html` (besides `style.css`) and check the result
7. In `style.css`, progressively add the style definitions of the following points; also, add the corresponding `<div>` and `<a>` elements at the beginning of the `<article>` section
8. Rollover button with transition:

```

#rolltr a.tr {
  font-weight: bold;
  padding: 5px 10px;
  background: #00ff00;
  box-shadow: 3px 3px 4px #999;
  border-radius: 20px;
  transition: background 1.0s ease, color 1.0s linear;
}

#rolltr a.tr:hover {
  background: #ffff00;
  color: #ff0000;
  box-shadow: 6px 6px 8px #999;
}

```

```

<br>
<div id="rolltr">
  <a href="#" class="tr">Press me!</a>
</div>

```

9. Other transition (box width change):

```

#tr2 {
  width: 100px;
  height: 100px;
  background: red;
  transition: width 2s;
  box-shadow: 6px 6px 8px #999;
}

#tr2:hover {
  width: 300px;
}

```

```
<br>
<div id="tr2">
  What do you think about this effect?
</div>
```

10. From <https://fonts2u.com/search.html?q=Kingthings+Christmas>, download the font “Kingthings Christmas” by clicking the button *@font-face* on the right (also in the new page that is opened). Alternatively, download another font from this or any other website offering free fonts. Save the zip file in the same folder as `index.html` and unzip the three files with the extensions `.svg`, `.ttf`, and `.woff`. Then use the font for the content of the `<div>` with `id="myfont"` (you can copy and paste the following code from <https://vision.unipv.it/wmt/lab/html5/cssfonts.txt>):

```
@font-face {
  font-family: "Kingthings Christmas";
  src: url("Kingthings_Christmas.woff") format("woff"),
       url("Kingthings_Christmas.ttf") format("truetype"),
       url("Kingthings_Christmas.svg#KingthingsChristmas")
       format("svg");
  font-weight: normal;
  font-style: normal;
}

#myfont {
  font-family: "Kingthings Christmas";
  font-size: 3em;
}
```

```
<br>
<div id="myfont">
  In the end, thanks to CSS3, fonts can be treated as external
  resources!
</div>
```

11. Two-dimensional transformation with transition:

```
#tr2d {
  width: 200px;
  height: 100px;
  background-color: yellow;
  transition: all 2s;
}

#tr2d:hover {
  transform: rotate(45deg);
}
```

```
<br>
<div id="tr2d">What do you think about this effect?</div>
```

12. Another rollover button:

(download to your folder the file <https://vision.unipv.it/wmt/lab/html5/tbl.jpg>)

```
#tbl {
  background-image: url("tbl.jpg");
  background-position: 50% 50%;
  background-repeat: no-repeat;
  background-origin: border-box;
  box-sizing: border-box;
  display: block;
  width: 86px; height: 86px;
  border: 43px solid rgba(0, 0, 0, 0.7);
  border-radius: 100%;
  transition: 0.5s ease;
}

#tbl:hover {
  border-width: 0px;
}
```

```
<br>
<a href=" https://www.w3.org/People/Berners-Lee/" id="tbl"></a>
```